

SASTRA DEEMED UNIVERSITY
(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

Dec 2023

Course Code: MAT102R01

Course: ENGINEERING MATHEMATICS - II

QP No. :UD135-A

Duration: 3 hours

Max. Marks:100

PART - A

Answer all the questions

10 x 2 = 20 Marks

1. Find the radius of curvature at any point $(0, c)$ of the catenary $y = c \cosh\left(\frac{x}{c}\right)$.
2. Define evolutes and involutes.
3. Write the Exact equation.
4. Write the Clairaut's equation.
5. If $\phi = xyz$, then find $\nabla\phi$.
6. If $\phi = \log(x^2 + y^2 + z^2)$, then find $\nabla\phi$.
7. Write the formula for directional derivative.
8. Define the surface integral and write the formula.
9. State the physical interpretation of the line integral $\int_A^B \vec{F} \cdot d\vec{r}$.
10. Write the Stokes theorem formula.

PART - B

Answer all the questions

4 x 15 = 60 Marks

11. a) Show that the radius of curvature at $(a, 0)$ on the curve $xy^2 = a^2(a - x)$ is $\frac{a}{2}$. (8)
- b) Find the equation of the circle of curvature of the parabola $y^2 = 12x$ at the point $(3, 6)$. (7)

(OR)

12. a) Find the minimum or maximum values of $3x^2 - y^2 + x^3$. (8)
- b) Find the envelop of $(x - a)^2 + (y - a)^2 = 2a$. (7)
13. a) Solve $(D^2 - 5D + 6)y = e^{5x}$. (8)
- b) A voltage Ee^{-at} is applied at $t = 0$ to a circuit containing inductance 'L' and resistance 'R'. Show that the current at any time 't' is $\frac{E}{R - aL}(e^{-at} - e^{-Rt/L})$ assuming that 'i=0' at $t = 0$. (7)

(OR)

14. Solve $(D^2 - 4D)y = e^x x$.
15. Find the equation of the tangent plane and normal line to the surface $xyz = 4$ at the point $\vec{i} + 2\vec{j} + 2\vec{k}$.

(OR)

16. a) Find 'a' and 'b' such that the surfaces $ax^2 - byz = (a + 2)x$ and $4x^2y + z^3 = 4$ cut orthogonally at $(1, -1, 2)$. (8)
- b) Find the angle between the surfaces $z = x^2 + y^2 - 3$ and $x^2 + y^2 + z^2 = 9$ at $(2, -1, 2)$. (7)

17. Verify Green's theorem in the XY plane for $\int_C \{(xy + y^2) dx + x^2 dy\}$, where 'C' is the closed curve of the region bounded by $y=x$ and $y=x^2$.

(OR)

- ✓ 18. Verify the Stokes theorem for a vector field defined by $F = (x^2 - y^2) \hat{i} - 2xy \hat{j}$, in the rectangular region in the XOY plane bounded by the lines $x=0$, $x=a$, $y=0$ and $y=b$.

PART - C

Answer the following

1 x 20 = 20 Marks

19. a) Solve $(3x + 2)^2 y'' + 3(3x + 2)y' - 36y = 3x^2 + 4x + 1$. (10)
- b) Verify the Gauss Divergence theorem for the vector $F = 4xz \hat{i} - y^2 \hat{j} + yz \hat{k}$ over the cube bounded by $x=0$, $x=1$, $y=0$, $y=1$, $z=0$, $z=1$. (10)
